



Figure 1: ROAR Logo

ROAR Installation Instructions

Requirements

- **Shell:** csh/tcsh (C Shell)
- **Make** build tool *## Installation Steps*

0. Ensure you are running csh/tcsh

This software requires the C Shell (csh) or tcsh. Verify your shell or switch to csh:

```
echo $SHELL
# If not running csh, switch to it:
csh
```

1. Download and Extract

Download and extract the ROAR software to the directory where you would like to install it:

```
tar -xzf roar.tar.gz -C /path/to/install/location
cd /path/to/install/location/roar
```

2. Build and Install

In the top-level **roar** directory, run Make to build and install the software:

```
make
```

This will install all required dependencies and create the **roar_env.csh** environment configuration file.

3. Set Up Environment

Before running the software, source the environment configuration file to set the required environment variables:

```
source roar_env.csh
```

Note: You will need to run this command each time you open a new terminal session, or add it to your `.cshrc` file for automatic loading:

```
echo "source /path/to/roar/roar_env.csh" >> ~/.cshrc
```

4. Run ROAR

Once the environment is set up, run the ROAR executable:

```
$ROAR_HOME/bin/roar
```

Troubleshooting

Environment Variables Not Set

If you receive errors about missing environment variables, ensure you have sourced the environment file:

```
source roar_env.csh
```

Permission Denied

If you encounter permission issues with the executable:

```
chmod +x $ROAR_HOME/bin/roar
```

Quick Start Summary

```
# 1. Extract and navigate to roar directory  
cd roar
```

```
# 2. Build  
make
```

```
# 3. Set environment  
source roar_env.csh
```

```
# 4. Run  
$ROAR_HOME/bin/roar
```

Usage

Operating Modes

ROAR operates in two primary modes:

- **Design Eqs Mode:** Work with design equations and circuit parameters
- **Device Params Mode:** Configure and adjust device-specific parameters

Toggle between modes using the interface controls or keyboard shortcuts.

Hot Keys

Plot Controls

Key	Action
x	Toggle X-axis log scale
y	Toggle Y-axis log scale
l	Toggle log scale (both axes)
v	Add/toggle vertical marker
h	Add/toggle horizontal marker

General

Key	Action
Ctrl+S	Save current state
Ctrl+O	Open/load saved state
Ctrl+Z	Undo last action
Ctrl+Y	Redo action
Ctrl+Q	Quit ROAR

Saving and Loading States

To save your current session: - Use Ctrl+S or File → Save State - States are saved to \$ROAR_HOME/states/ by default

To load a previously saved state: - Use Ctrl+O or File → Load State - Navigate to your saved .roar state file

Tips

- Save states frequently when working on complex designs
- Use descriptive filenames for saved states (e.g., `amplifier_v2.roar`)
- Switch to Device Params mode to fine-tune component values after initial design

Known Issues

- 3-D Plotting functionality is currently limited and buggy; use 2-D plots for best results.